Chapter 3
Boundary-Value Problems in Electrostatics: II

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These notes follow the notation and organization of Jackson's Chapter 3. I kept the emphasis on the structure of the differential equations, the role of orthogonal functions, and the way Green functions adapt to separable geometries. The aim here is not to reproduce every algebraic line from the book, but to keep the logic of the chapter visible enough that the method behind each result is hard to lose.

Prepared from the uploaded Chapter 3 notes PDF.

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How this chapter fits into the story

Chapter 1 gave the general field equations and Green's theorem. Chapter 2 showed how boundary conditions can sometimes be handled almost geometrically, especially by images. Chapter 3 is where the orthogonal-function machinery really takes over. The central idea is simple: if the boundary matches a coordinate system in which Laplace's equation separates, then the natural basis functions of that coordinate system become the language of the solution. In spherical problems this means Legendre functions and spherical harmonics; in cylindrical problems it means Bessel functions; in both cases the Green function can be expanded in the same basis.

1. Laplace Equation in Spherical Coordinates

In spherical coordinates (r, θ, ϕ) the Laplace equation is

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}(r\Phi) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\Phi}{\partial\phi^2} = 0.$$

Jackson writes the separated form as

$$\Phi(r,\theta,\phi) = \frac{U(r)}{r}P(\theta)Q(\phi).$$

Substitution isolates the azimuthal part first, so one sets

$$\frac{1}{Q}\frac{d^2Q}{d\phi^2} = -m^2, \qquad Q = e^{\pm im\phi}.$$

Single-valuedness under $\phi \to \phi + 2\pi$ forces m to be an integer. The remaining angular and radial equations are

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) + \left[l(l+1) - \frac{m^2}{\sin^2\theta}\right]P = 0,$$

$$\frac{d^2U}{dr^2} - \frac{l(l+1)}{r^2}U = 0.$$

The radial equation is already telling: it is scale-invariant, so its solutions are pure powers,

$$U = Ar^{l+1} + Br^{-l}.$$

Therefore the potential itself has the characteristic spherical form

$$\Phi \sim Ar^l + Br^{-(l+1)}.$$

This is one of the structural facts of electrostatics worth memorizing. In spherical geometry, regular solutions near the origin are positive powers of r, while decaying solutions outside sources are inverse powers.

2. Legendre Equation and Legendre Polynomials

For $m = 0$ it is convenient to set $x = \cos\theta$. Then the angular equation becomes

$$\frac{d}{dx}\left[(1-x^2)\frac{dP}{dx}\right] + l(l+1)P = 0,$$

which is the ordinary Legendre equation. Jackson develops the solution by power series and shows that finiteness at $x = \pm 1$ forces the series to terminate. That restriction is what quantizes l to a nonnegative integer and turns the regular solutions into the Legendre polynomials $P_l(x)$.

The first few are

$$P_0(x) = 1, \qquad P_1(x) = x, \qquad P_2(x) = \frac{1}{2}(3x^2 - 1),$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x), \qquad P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3).$$

A compact representation is Rodrigues' formula,

$$P_l(x) = \frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l.$$

The orthogonality relation is

$$\int_{-1}^{1} P_l(x)P_{l'}(x)\,dx = \frac{2}{2l+1}\,\delta_{ll'}.$$

This is the statement that matters for boundary-value problems. Once the potential on a spherical boundary is known as a function of θ, the coefficients in the interior or exterior solution are determined by projection onto the P_l basis. In other words, separation of variables by itself gives a family of solutions; orthogonality is what turns that family into a calculational tool.

Jackson also writes a Legendre-series expansion for a generic function,

$$f(x) = \sum_{l=0}^{\infty} A_l P_l(x), \qquad A_l = \frac{2l+1}{2}\int_{-1}^{1} f(x)P_l(x)\,dx.$$

That formula is the spherical analogue of a Fourier series, and it is used repeatedly in the rest of the chapter.

3. Boundary-Value Problems with Azimuthal Symmetry

If a problem has azimuthal symmetry, then $m = 0$ and the general solution simplifies to

$$\Phi(r,\theta) = \sum_{l=0}^{\infty}\left(A_l r^l + B_l r^{-(l+1)}\right)P_l(\cos\theta).$$

This is the natural expansion for all axisymmetric spherical problems. The geometry determines whether the A_l or B_l branch survives. For example:

- inside a sphere containing no source at the origin, one keeps only r^l;
- outside a localized source region, one keeps only $r^{-(l+1)}$ if the potential must vanish at infinity.

If the potential is specified on a sphere $r = a$ as $V(\theta)$, then for the interior problem

$$V(\theta) = \sum_{l=0}^{\infty} A_l a^l P_l(\cos\theta),$$

so that

$$A_l = \frac{2l+1}{2a^l} \int_0^{\pi} V(\theta) P_l(\cos\theta) \sin\theta \, d\theta.$$

The exterior coefficients are obtained in the same way, with a^l replaced by $a^{-(l+1)}$.

A point that Jackson emphasizes and that is easy to overlook is uniqueness of representation. Once the proper regularity conditions are imposed, the coefficients are not arbitrary ornaments. They encode the full physics, and each term has a recognizable multipole meaning. The $l = 0$ term is monopolar, $l = 1$ dipolar, $l = 2$ quadrupolar, and so on.

The same section gives a very useful expansion of the Coulomb kernel:

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \sum_{l=0}^{\infty} \frac{r_<^l}{r_>^{l+1}} P_l(\cos\gamma),$$

where $r_<$ and $r_>$ are the smaller and larger of r and r', and γ is the angle between $\mathbf{x}$ and $\mathbf{x}'$. This identity is one of the true workhorses of the subject. It is the bridge between local spherical harmonics and the nonlocal Green kernel.

4. Behavior of Fields in a Conical Hole or Near a Sharp Point

This section is one of the most physical in the chapter. One considers an axisymmetric region bounded by a cone $\theta = \beta$ held at fixed potential. The angular equation is no longer required to be regular on the whole interval $-1 \le x \le 1$; only the restricted range $\cos\beta \le x \le 1$ matters. The result is that the allowed separation parameter need not be an integer. Jackson writes it as ν, and the regular separated solutions become

$$\Phi(r, \theta) \sim A r^\nu P_\nu(\cos\theta),$$

with the eigenvalue condition

$$P_\nu(\cos\beta) = 0.$$

This changes the near-origin behavior dramatically. The dominant field components scale as

$$E_r \sim r^{\nu-1}, \qquad E_\theta \sim r^{\nu-1}.$$

Hence the geometry decides whether the field is suppressed or singular near the tip:

- for a narrow conical hole ($\beta < \pi/2$), one has $\nu > 1$, so the field goes to zero near the vertex;
- for a sharp conducting point ($\beta > \pi/2$), one has $0 < \nu < 1$, so the field diverges as $r \to 0$.

This is a clean mathematical explanation of field enhancement near sharp tips. The singularity is not mysterious; it is the inevitable consequence of the lowest allowed exponent in the separated solution.

Jackson also gives useful approximations for ν in extreme limits. For small β,

$$\nu \approx \frac{2.405}{\beta} - \frac{1}{2},$$

while for $\pi - \beta \ll 1$,

$$\nu \approx \left[2 \ln\left(\frac{2}{\pi - \beta}\right)\right]^{-1}.$$

The second form shows just how slowly ν tends to zero as the point becomes very sharp.

5. Associated Legendre Functions and the Spherical Harmonics $Y_{lm}(\theta, \phi)$

When azimuthal symmetry is absent, $m \neq 0$, and the relevant angular functions are the associated Legendre functions. For positive m,

$$P_l^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_l(x).$$

A more symmetric formula valid in either sign convention is obtained from Rodrigues' representation. The key relation for negative m is

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x).$$

These functions are orthogonal in l for fixed m,

$$\int_{-1}^{1} P_l^m(x) P_{l'}^m(x)\, dx = \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \delta_{ll'}.$$

Combining the θ- and ϕ-dependence leads to the normalized spherical harmonics,

$$Y_{lm}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}}\, P_l^m(\cos\theta) e^{im\phi}.$$

They satisfy

$$\int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta\, d\theta\, Y_{l'm'}^* Y_{lm} = \delta_{ll'} \delta_{mm'}.$$

This is the natural orthonormal basis on the sphere. In practice, every well-behaved angular function

can be expanded as

$$g(\theta,\phi)=\sum_{l=0}^{\infty}\sum_{m=-l}^{l}A_{lm}Y_{lm}(\theta,\phi),\qquad A_{lm}=\int d\Omega\, Y_{lm}^{*}(\theta,\phi)g(\theta,\phi).$$

Once this basis is available, the fully general solution of Laplace's equation in spherical coordinates can be written as

$$\Phi(r,\theta,\phi)=\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\left(A_{lm}r^{l}+B_{lm}r^{-(l+1)}\right)Y_{lm}(\theta,\phi).$$

That formula is the complete spherical counterpart of the Fourier expansion of a two-dimensional potential.

6. Addition Theorem for Spherical Harmonics

The addition theorem is one of the central identities in the chapter:

$$P_l(\cos\gamma)=\frac{4\pi}{2l+1}\sum_{m=-l}^{l}Y_{lm}^{*}(\theta',\phi')Y_{lm}(\theta,\phi),$$

where γ is the angle between the two directions (θ,ϕ) and (θ',ϕ').

This theorem deserves to be understood, not only remembered. The left-hand side depends only on the relative angle between two directions; the right-hand side rewrites that rotationally invariant quantity as a bilinear combination of basis functions defined with respect to a fixed coordinate system. It is the precise statement that spherical harmonics provide the irreducible angular content of the Coulomb kernel.

A direct consequence is the fully factorized form of the free-space Green kernel:

$$\frac{1}{|\mathbf{x}-\mathbf{x}'|}=4\pi\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\frac{1}{2l+1}\frac{r_{<}^{l}}{r_{>}^{l+1}}Y_{lm}^{*}(\theta',\phi')Y_{lm}(\theta,\phi).$$

This identity is what turns many charge-density integrals into manageable radial integrals plus angular projections.

7. Laplace Equation in Cylindrical Coordinates; Bessel Functions

In cylindrical coordinates (ρ,ϕ,z),

$$\frac{\partial^2\Phi}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial\Phi}{\partial\rho}+\frac{1}{\rho^2}\frac{\partial^2\Phi}{\partial\phi^2}+\frac{\partial^2\Phi}{\partial z^2}=0.$$

With the product ansatz

$$\Phi(\rho,\phi,z)=R(\rho)Q(\phi)Z(z),$$

one finds

$$\frac{d^2Z}{dz^2} - k^2 Z = 0, \qquad \frac{d^2Q}{d\phi^2} + \nu^2 Q = 0,$$

$$\frac{d^2R}{d\rho^2} + \frac{1}{\rho}\frac{dR}{d\rho} + \left(k^2 - \frac{\nu^2}{\rho^2}\right) R = 0.$$

After setting $x = k\rho$, the radial equation becomes Bessel's equation,

$$\frac{d^2R}{dx^2} + \frac{1}{x}\frac{dR}{dx} + \left(1 - \frac{\nu^2}{x^2}\right) R = 0.$$

Its regular power-series solution is the Bessel function of the first kind,

$$J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{j=0}^{\infty} \frac{(-1)^j}{j!\,\Gamma(j+\nu+1)} \left(\frac{x}{2}\right)^{2j}.$$

The second linearly independent solution is the Neumann function $N_\nu(x)$, and the outgoing/incoming wave combinations are the Hankel functions,

$$H_\nu^{(1)}(x) = J_\nu(x) + iN_\nu(x), \qquad H_\nu^{(2)}(x) = J_\nu(x) - iN_\nu(x).$$

The limiting forms matter constantly in practice:

$$J_\nu(x) \sim \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu, \qquad x \ll 1,$$

$$J_\nu(x) \sim \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right), \qquad x \gg 1.$$

Jackson then proves the orthogonality relation for Bessel functions whose arguments are scaled by the zeros $x_{\nu n}$ of J_ν:

$$\int_0^a \rho J_\nu\left(x_{\nu n}\frac{\rho}{a}\right) J_\nu\left(x_{\nu n'}\frac{\rho}{a}\right) d\rho = \frac{a^2}{2}[J_{\nu+1}(x_{\nu n})]^2 \delta_{nn'}.$$

This leads to the Fourier–Bessel expansion

$$f(\rho) = \sum_{n=1}^{\infty} A_{\nu n} J_\nu\left(x_{\nu n}\frac{\rho}{a}\right),$$

with coefficients determined by projection. So again the pattern is the same as before: separation gives the basis, orthogonality makes the basis useful.

Jackson also notes the modified Bessel equation obtained when the separation constant has the opposite sign. The corresponding solutions are $I_\nu(x)$ and $K_\nu(x)$. These become indispensable whenever exponential growth or decay in the radial direction is the physically relevant behavior.

8. Boundary-Value Problems in Cylindrical Coordinates

For a finite cylinder of radius a and height L, with the side and bottom grounded and the top held at $\Phi = V(\rho, \phi)$, the separated solution takes the form

$$\Phi(\rho, \phi, z) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} J_m(k_{mn}\rho) \sinh(k_{mn}z) \left(A_{mn} \sin m\phi + B_{mn} \cos m\phi\right),$$

where

$$k_{mn} = \frac{x_{mn}}{a}, \qquad J_m(x_{mn}) = 0.$$

The coefficients are fixed by expanding the boundary data $V(\rho, \phi)$ in a Fourier series in ϕ and a Fourier–Bessel series in ρ.

What matters conceptually here is that cylindrical boundary conditions select zeros of Bessel functions in exactly the same way that rectangular boxes select integer multiples of π and spherical boundaries select Legendre coefficients. The eigenvalues are the geometric fingerprints of the coordinate system.

Jackson also discusses the infinite half-space $z \geq 0$, where sums over discrete radial modes are replaced by integrals over continuous k. In that limit one arrives naturally at Hankel transforms. The discrete Fourier–Bessel series turns into an integral representation, just as ordinary Fourier series turn into Fourier transforms on unbounded intervals.

9. Expansion of Green Functions in Spherical Coordinates

The same basis that solves Laplace's equation also expands the Green function satisfying the appropriate spherical boundary conditions. In free space one has

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = 4\pi \sum_{l,m} \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi).$$

For a spherical boundary at $r = a$, the Green function is obtained by adding a homogeneous term chosen so that the correct boundary condition is enforced. For the exterior Dirichlet problem, Jackson writes a series of the schematic form

$$G(\mathbf{x}, \mathbf{x}') = 4\pi \sum_{l,m} \frac{1}{2l+1} \left[\frac{r_<^l}{r_>^{l+1}} - (\text{image-like correction}) \right] Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi).$$

The exact correction is determined by demanding $G = 0$ at the boundary. The important lesson is that the image method and the orthogonal-function method are not competing viewpoints here; they are two faces of the same Green function.

Once the series for G is known, Poisson problems in spherical geometry reduce to radial integrals plus angular harmonic projections. This is one of the cleanest demonstrations in the book of why the Green-function formalism becomes especially transparent in separable coordinates.

10. Solution of Potential Problems with Spherical Green Function Expansion

After the Green function has been expanded, the next step is to use it rather than admire it. The potential follows from Jackson's general representation,

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int_V \rho(\mathbf{x}')G(\mathbf{x}, \mathbf{x}')\, d^3x' + \text{boundary term}.$$

If the boundary data live naturally on a sphere, one expands both the source and the boundary function in spherical harmonics. The angular integrations become diagonal because of orthogonality. What remains is usually a radial integral for each (l, m) channel.

This way of organizing the calculation is extremely efficient. Instead of solving one three-dimensional problem all at once, one solves an infinite family of one-dimensional radial problems, each labeled by angular momentum indices. The decomposition is exact, not approximate, and the truncation error is completely under control when the source has simple angular structure.

11. Expansion of Green Functions in Cylindrical Coordinates

In cylindrical geometries the same story repeats with a different basis. The free-space Green function can be written as a sum or integral over m and k, with angular dependence $e^{im(\phi-\phi')}$, axial dependence $e^{-k|z-z'|}$ or its hyperbolic variants, and radial dependence expressed through Bessel or modified Bessel functions.

In schematic form one encounters representations like

$$G(\mathbf{x}, \mathbf{x}') \sim \sum_{m=-\infty}^{\infty} \int_0^\infty dk\, \mathcal{R}_m(k; \rho, \rho') e^{im(\phi-\phi')} e^{-k|z-z'|},$$

where $\mathcal{R}_m$ is built from the appropriate cylindrical solutions and then modified to satisfy the boundary conditions.

The conceptual point is the same as in the spherical case, but the bookkeeping is different: cylindrical symmetry does not quantize the axial variable on an unbounded interval, so integrals over k appear naturally. One gets a mixed discrete–continuous spectral representation.

12. Eigenfunction Expansions for Green Functions

This section makes the underlying principle explicit. A Green function is not just a special trick cooked up from an image or guessed from experience. It can be built systematically from the eigenfunctions of the self-adjoint differential operator associated with the boundary-value problem.

If $\{u_n\}$ are orthonormal eigenfunctions satisfying

$$\mathcal{L}u_n = \lambda_n u_n,$$

then the Green function is represented schematically as

$$G(\mathbf{x}, \mathbf{x}') = \sum_n \frac{u_n(\mathbf{x}) u_n^*(\mathbf{x}')}{\lambda_n},$$

with the obvious modifications when part of the spectrum is continuous.

For electrostatics the operator is essentially the Laplacian together with the imposed boundary conditions. This eigenfunction viewpoint is the abstract version of everything Jackson has been doing concretely with Legendre, Bessel, and Fourier bases. It is useful because it shows what all those expansions have in common and why they must work.

13. Mixed Boundary Conditions, Conducting Plane with a Circular Hole

The final section addresses a problem that is geometrically simple but methodologically richer: a conducting plane with a circular aperture. One does not have a pure Dirichlet or pure Neumann condition everywhere on the boundary. Instead, different pieces of the boundary obey different conditions. This is the natural setting for mixed boundary conditions.

The main lesson is that mixed problems are still linear, but the basis functions and the matching conditions become subtler. One cannot impose a single convenient boundary condition globally on the whole surface. Instead one constructs the solution piece by piece, usually with an integral transform or a specially adapted Green function, and then matches the conditions across the different boundary segments.

For the circular hole, cylindrical coordinates are unavoidable, and the result is naturally expressed in terms of Bessel-function integrals. Physically the solution describes the fringing of electric field lines near the rim of the aperture. Mathematically it is a reminder that separability alone is not the whole story: one also needs to know how the imposed data sit on the boundary.

Closing synthesis

The intellectual center of Chapter 3 is not any one special function. It is the idea that geometry chooses the basis. Spherical boundaries lead to Legendre functions and spherical harmonics; cylindrical boundaries lead to Bessel functions; Green functions inherit the same eigenfunction structure. Once that is understood, many formulas that first look unrelated fall into place. The chapter is really teaching one habit of mind: never attack a boundary-value problem before asking which coordinate system makes the boundary look natural and which orthogonal family belongs to that coordinate system.

That is the point where electrostatics starts to feel less like a list of solved problems and more like a coherent mathematical method.